

2026 Summer Project Work

MDP PRISM | KAN-PINN + T-KAN

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2026 MDP Summer Fellowship Cohort

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Project Plan

Description of Work

This project addresses milling chatter, a self-excited vibration in machining that limits productivity and damages tools when left unchecked. Milling chatter dynamics follow a regenerative delay differential equation in which the cutting force at time t depends on the tool position at time t minus tau, with tau set by the spindle speed and the number of teeth. Predicting and explaining chatter from sensor signals is a core problem in advanced manufacturing. Existing approaches either give black-box predictions without physical insight, or rely on hand-derived equations that miss real-world nonlinearities.

My summer work builds the first Temporal Kolmogorov-Arnold Network Physics-Informed Neural Network (T-KAN-PINN) for these regenerative dynamics, applied to the Digital Machining Database benchmark of 120 simulated milling cases with known ground-truth physical parameters. T-KAN extends the standard KAN architecture to handle sequential time-series windows, making it better suited for the delay-embedded state space of the milling DDE than a static KAN. The contribution sits at Phase 5 of the broader MDP PRISM pipeline, where Prof. Abell positioned the KAN-PINN as the primary scientific vehicle for physics extraction. Phases 1 and 2, covering feature engineering, EDA, and an XGBoost classifier with SHAP, PDP, LIME, and DiCE explainers, finished during the prior semester (OOF AUC 0.968, F1 0.901 under leave-dataset-out validation). Phase 4 (SINDy-DDE) confirmed that the regenerative chip thickness term discriminates stable from unstable cuts across all 120 datasets, but collapses under 5% sensor noise, directly motivating the T-KAN-PINN.

Seven interpretability and analysis tools wrap the T-KAN-PINN to make every layer of the model interrogable: power spectral density analysis to characterise frequency-domain chatter signatures, KL divergence to quantify distribution shift between stable and unstable regimes and act as a physics-aware training loss, PySR symbolic regression on learned T-KAN edges to extract closed-form physics, input sensitivity mapping over physical parameters, SHAP rankings on PINN inputs, DiCE counterfactuals on machinist-controllable variables (omega and b), and Monte Carlo drift testing for risk-rated recommendations. The decisive advantage of working with the Digital Machining Database is that every simulation has known ground-truth parameters (ktc, fn, damping, teeth, tau). Most PINN papers validate on toy problems or field data without ground truth, so this benchmark structure lets the work make quantitative claims about parameter recovery, stability boundary reconstruction, and symbolic extraction fidelity. The required deliverable is a technical report for Prof. Abell. The stretch deliverable is a NeurIPS Machine Learning and Physical Sciences workshop paper.

Summer Goals

The work breaks into nine phases. Phase 1 covers setup and literature. Phase 2 covers power spectra and KL divergence analysis. Phases 3 to 5 cover model construction (core T-KAN-PINN, conditional architecture across 60 datasets, inverse parameter inference). Phases 6 to 7 cover physics extraction and robustness (PySR symbolic regression, multi-fidelity noise training). Phase 8 covers ablations and counterfactual XAI. Phase 9 covers writeup.

Required deliverable

A technical report by August 14, 2026, documenting the KAN-PINN architecture, training procedure, validation results, symbolic extraction, and full XAI findings. Delivered to Prof. Abell.

Stretch deliverable

A NeurIPS ML4PS workshop paper draft for the early-September submission window. Backup venues with later cycles include ICML AI4Science, AAAI AI4Science, and ASME IDETC and MSEC.

Quantitative success criteria

Inverse-problem accuracy within 10% of ground-truth ktc, fn, and damping on at least 80% of held-out datasets. Reconstructed stability lobes within 15% of the analytical Altintas-Budak prediction across the operating range. Symbolic expression with R squared above 0.90 against the validated DDE structure on clean data. KL divergence between stable and unstable PSD profiles significantly higher near b/blim = 1.0 than far from it. Per-method noise robustness curves at 0%, 1%, and 5% sensor noise tiers for T-KAN-PINN, PySR, and KL divergence. Full ablation matrix covering T-KAN versus MLP, KL divergence loss on or off, residual physics on or off, hardcoded versus learnable delay. Clear DiCE counterfactuals on at least 10 representative unstable operating points, each with Monte Carlo survival rates above 0.85.

Qualitative success criteria

The recovered symbolic expression contains the regenerative term x(t) minus x(t minus tau) without explicit hardcoding. PINN-based SHAP rankings agree directionally with classical chatter theory (ktc and b drive instability up, damping and fn drive stability up). Hidden-layer activations encode physically meaningful quantities like tooth-pass frequency or fn proximity, verifiable via correlation analysis. The narrative is coherent enough to circulate to the parallel subteams within MDP PRISM.

Schedule of Tasks & Sub-Tasks

The schedule budgets 300 hours across 15 weeks, capped at 20 hours per week. The full week-by-week schedule with start dates, end dates, and per-task hours lives in the accompanying Project_Schedule.xlsx file. The summary table below shows the nine phases at a glance.

Phase 2 (power spectra and KL divergence) overlaps with the start of Phase 3 (T-KAN-PINN core) since the KL divergence output feeds directly into the stability-aware training loss. Phase 5 (inverse problem) starts in the last few days of Phase 4 (conditional forward), since the inverse pipeline reuses the trained conditional model. Phase 6 (PySR) starts as Phase 5 wraps, since extracting symbolic expressions from T-KAN edges does not require the inverse pipeline.

#	Phase	Hours	Window
1	Setup, literature, infrastructure	20	May 1 to May 14
2	Power spectra and KL divergence analysis	25	May 15 to May 28
3	Core T-KAN-PINN implementation	**70**	May 22 to June 18
4	Conditional forward problem across 60 datasets	**45**	June 12 to July 2
5	Inverse problem and parameter inference	35	June 29 to July 12
6	PySR symbolic extraction from T-KAN edges	35	July 13 to July 26
7	Noise robustness via multi-fidelity training	25	July 27 to August 2
8	Ablations and DiCE counterfactuals	25	August 3 to August 9
9	Technical report and workshop paper writeup	20	August 10 to August 15
	Total	**300**	**15 weeks at 20 hr per week**

Phase gates and decision points

Three checkpoints govern descope decisions during the summer. End of week 7 (June 18): single-dataset T-KAN-PINN must reproduce chatter on at least one stable and one unstable dataset. End of week 8 (June 25): the conditional model must hit per-dataset trajectory R squared above 0.85 on at least 40 of 60 clean datasets. End of week 12 (July 23): PySR must produce at least one extracted expression with the regenerative term structure recoverable. If a gate is missed, the project descopes per the contingency ladder in the Risks section.

Risks & Contingencies

Seven risks shape the contingency strategy. Each one has a mitigation plan and a fallback position.

R1. T-KAN-PINN training fails to converge

Probability: Medium. Impact: High. Weeks 3 to 6 are budgeted for debugging collocation density, delay initialization, and gradient instabilities. Training starts on a single clean dataset before scaling to all 60. Fallback: switch to MLP-PINN with post-hoc PySR symbolic regression. This pivot still yields a novel result for milling chatter and keeps the workshop paper viable.

R2. KL divergence loss destabilises training

Probability: Medium. Impact: Medium. The KL divergence term is introduced with a small initial weight and increased only after the data and physics losses converge. Fallback: use KL divergence as a post-hoc validation metric rather than a training loss. The power spectra and KL divergence results from Phase 2 are fully independent of T-KAN-PINN training and are always deliverable regardless of this risk.

R3. Inverse problem fails to recover ground-truth parameters

Probability: Medium-High. Impact: Medium. Known parameters are used as initialization for the conditional training, reducing cold-start risk. Testing starts on the cleanest single-dataset case first. Fallback: if adjoint-based inference is unstable, report the forward problem (trajectory reconstruction) and symbolic extraction results instead. This still supports a workshop paper, repositioned as a method paper rather than a parameter-inference paper.

R4. PySR extracts noisy or overcomplicated expressions

Probability: Medium. Impact: Medium. The operator library is restricted to physics-relevant primitives with a BIC parsimony penalty. Fallback: report numerical sensitivity analysis alongside partial symbolic structure. Compare against E-SINDy as an alternative baseline. If both methods produce messy expressions, the comparison itself becomes a finding.

R5. Great Lakes HPC compute time becomes a bottleneck

Probability: Low-Medium. Impact: Medium. Admin rights under the jaabell project are already provisioned via the ARC RMP portal. All training jobs use checkpointing every 100 epochs. Fallback: short-duration prototyping runs on Kaggle GPU instances, with only final long-form training jobs submitted to Great Lakes.

R6. pykan or T-KAN library limitations or bugs

Probability: Low. Impact: Medium. pykan is pinned to a stable release from the start. Fallback: a PyTorch-native KAN reimplementation using B-spline edges. This avoids any upstream dependency issues without changing the core architecture.

R7. Team coordination delays

Probability: Low. Impact: Low. The KAN-PINN work is independent of the parallel subteams. The only shared resource is the /nfs/turbo/coe-jaabell/PRI2M directory, which already holds the finalized feature CSVs. Weekly 30-minute check-ins with Prof. Abell at the start of each week surface any coordination issues before they affect the schedule.

Contingency ladder

If multiple risks hit together, the project descopes in this order. Tier 1 (full plan): all nine phases, technical report, and ML4PS draft. Tier 2 (forward only): drop the inverse problem, keep forward training, PySR extraction, power spectra, KL divergence, and ablations, and reframe the workshop paper as a method paper. Tier 3 (single-dataset): drop conditional training, demonstrate T-KAN-PINN on 4 to 8 curated datasets, and still deliver the technical report. Note that power spectra and KL divergence results from Phase 2 are always deliverable regardless of which tier is reached, since they are independent of T-KAN-PINN training. Tier 4 (architecture pivot): if pykan or T-KAN fails entirely, pivot to MLP-PINN with PySR; the existing feature pipeline already supports this.

Executive Summary/Conclusions

This section is to be completed at the end of the fellowship period (August 15, 2026). It will cover: what work was completed across all nine phases, which results were strongest across T-KAN-PINN training, power spectra and KL divergence analysis, PySR symbolic extraction, and noise robustness, where the approach ran into limitations, and what the next steps are for fall. It will also include highlights of the technical report and any open items for the broader MDP PRISM team.

In addition to writing about your research, please also answer the following reflection questions regarding the Multidisciplinary Design Program and Summer Program.

1. [Summer Program] Which weekly session(s) did you enjoy the most and why?
2. [Summer Program] Is there anything you wish we covered over the summer?
3. [MDP-all year] Has MDP been a valuable experience and if so, why? In what ways could MDP be better?